23PCCE501L
AIML Practical Examination

Disaster Management and Rescue Route Planning

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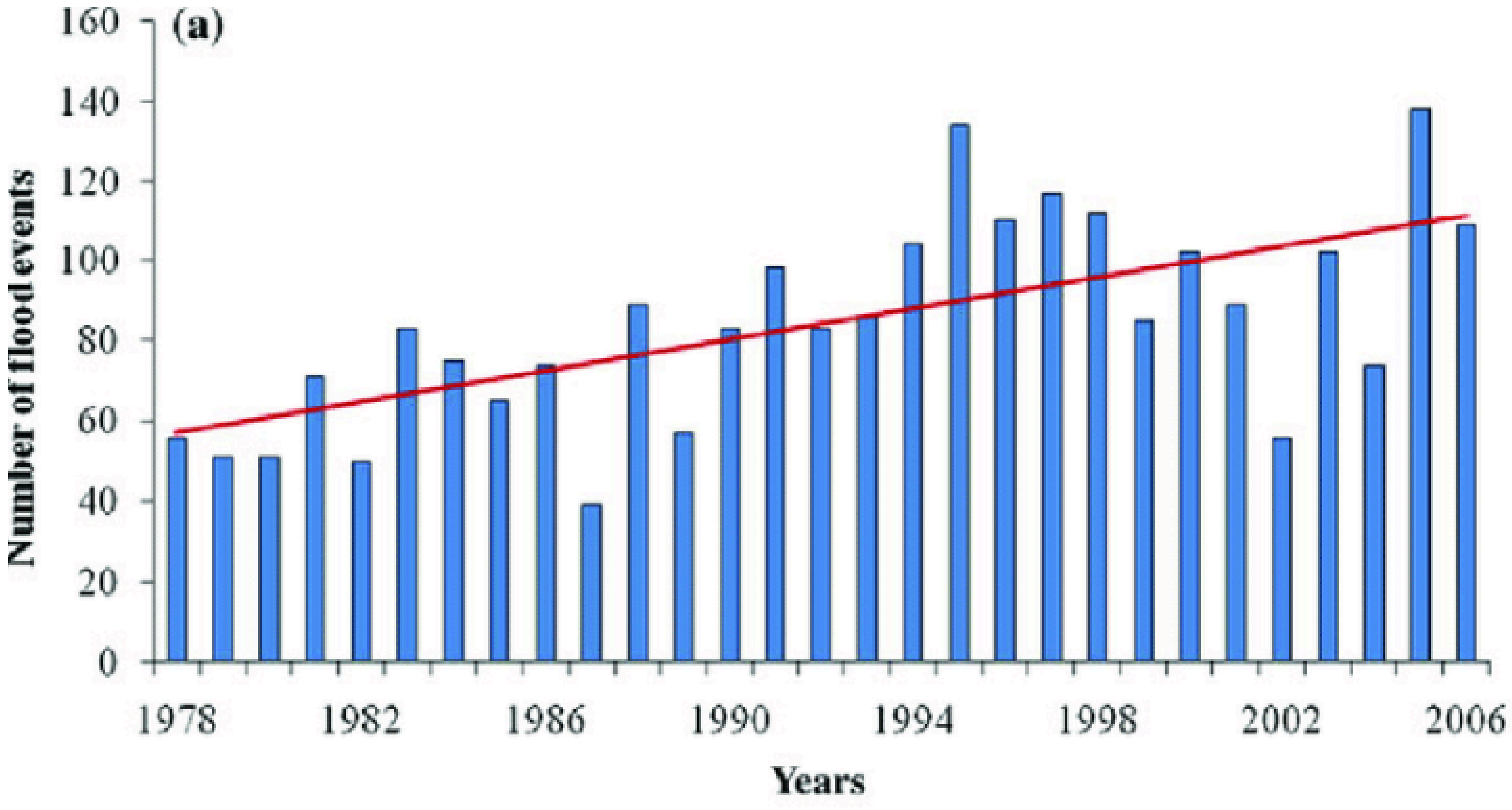
Introduction

Floods are one of the most common and destructive natural disasters. They become dangerous mainly due to:

- Sudden Onset: Flash floods can occur with little or no warning, leaving almost no time for preparation or evacuation.
- Widespread Damage: They destroy infrastructure, disrupt essential services, and threaten human lives.
- Post-Flood Diseases: Stagnant and contaminated floodwater often leads to outbreaks of waterborne diseases, increasing health risks even after the floods recede.



Traditional flood response systems rely on slow official communication and physical rescue deployment, which delays help when every second counts. Rising water, debris, and blocked roads further limit rescue movement, while victims struggle to identify safe evacuation routes or shelters as conditions change rapidly. Real-time information on water levels and safe paths is rarely available, people often make dangerous decisions in uncertainty. With smartphones being the most accessible tool people carry during disasters, mobile-based solutions become essential—providing instant alerts, real-time safety updates, and guided evacuation support that traditional methods cannot offer.

Data Collection

Datasetlink:
https://www.kaggle.com/datasets/s3programmer/flood-risk-in-india

This is the flood dataset which we have used, it has columns like:

Latitude, Longitude, water level, river discharge, rainfall, temperature, humidity, elevation, land cover, soil type, population, infrastructure, historical floods, flood occured.

It has 10,000 records.

About the system

This system takes live location of the user as input and then after clicking on SOS it tell the flood risk level in that area,nearest rescue center, displays shortest and safest path to reach the rescue center- displays it on map and also displays some safety tips.

Changes suggested:

- Predict number of ambulances required in an rescue center based on population
- Build a chatbot for user assistance

Methodology

The system combines unsupervised learning, supervised learning, and AI-based pathfinding to build a real-time Intelligent Flood Emergency Response System for India.

Data Preparation

1. Historical flood records are cleaned and filtered.
2. Key features (Latitude, Longitude, Water Level) are selected.
3. StandardScaler is applied .

Flood Risk Clustering

1. K-Means clustering is used to form four flood-risk zones: No, Low, Medium, High.
2. Clusters are formed based on similarity in water-level severity.
3. These cluster labels become the ground truth

Flood Risk Prediction

1. Logistic Regression is trained using the cluster-based labels.
2. The model predicts risk levels for any new or real-time latitude–longitude input.
3. Ensures fast and accurate flood risk classification during emergencies.

Methodology

Strategic Rescue Center Allocation

1. K-Means is applied again inside each risk zone to position rescue centers optimally.
2. A minimum separation distance is enforced to avoid clustering too many centers in the same area.
3. Ensures maximum regional coverage and efficient resource distribution.

Route Corridor Densification & Risk Evaluation

1. Extra intermediate points are generated along the path.
2. Each point is assigned a flood-risk level using the trained classifier.

Risk-Aware Pathfinding (A Algorithm)*

1. A KDTree speeds up nearest-neighbor searches for risk and distance checks.
2. A* algorithm is modified to account for both distance and flood risk.
3. Produces the safest and shortest evacuation route.

Real-Time Deployment Using Gradio

- SOS button automatically detects the user's GPS location.
  - Displays: Current location risk, Nearest rescue center, Safety tips, Interactive map with optimized safe route
- Enables rapid, data-driven decision-making during flood emergencies.

Methodology for change suggested

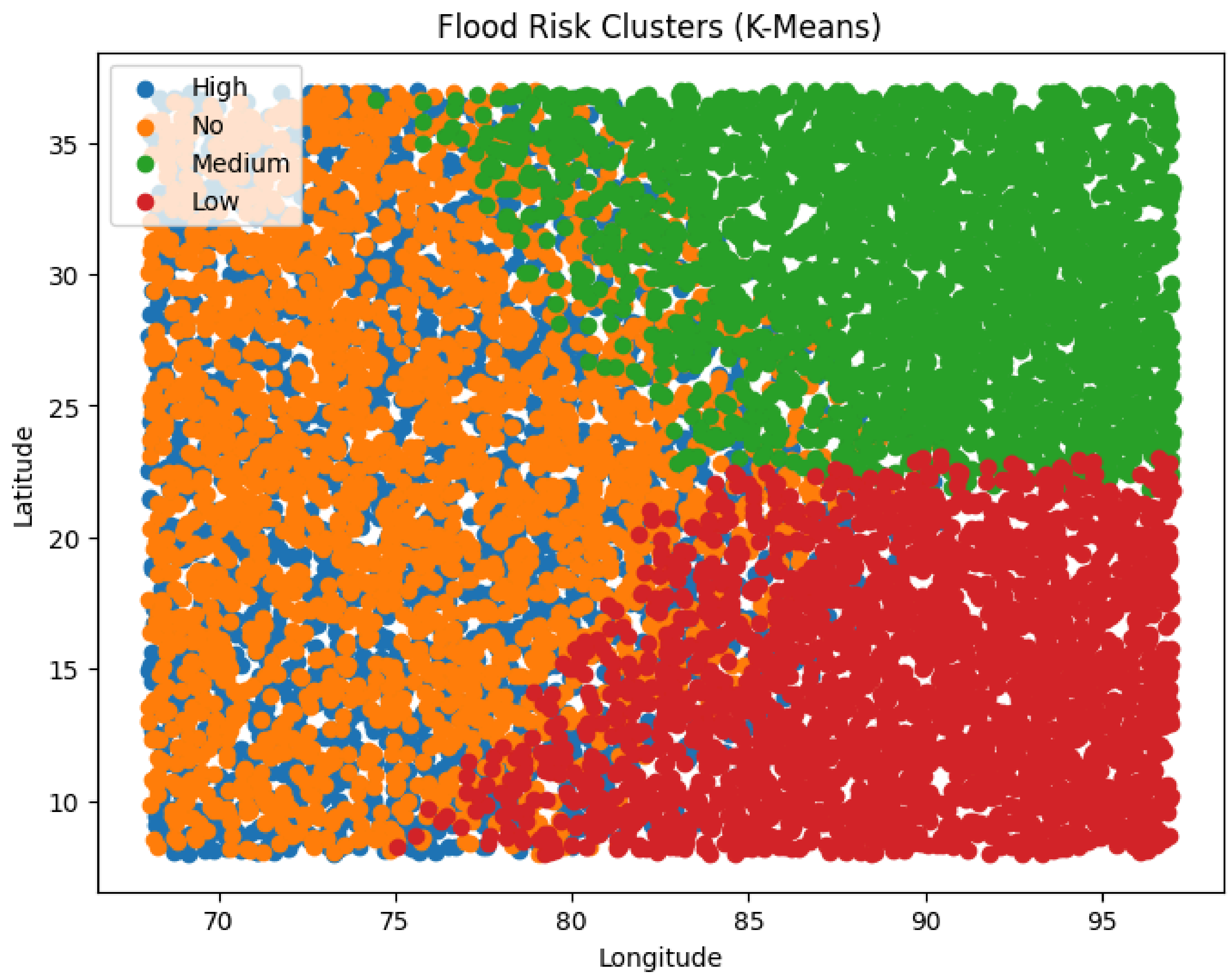
Fleet (Ambulance) Management:

- We have used random forest Classifier for fleet planning.
- Each center is initially given a fixed number of fleets.

Chatbot:

- The chatbot first uses NLP techniques to understand the user’s text—detecting intent, extracting keywords, and interpreting the meaning of the message.
- After extracting words it consoles the flood victim and directas it to move from that place.
- A trained AI model (like GPT) generates the response by predicting the most relevant and accurate text based on patterns learned from massive datasets.

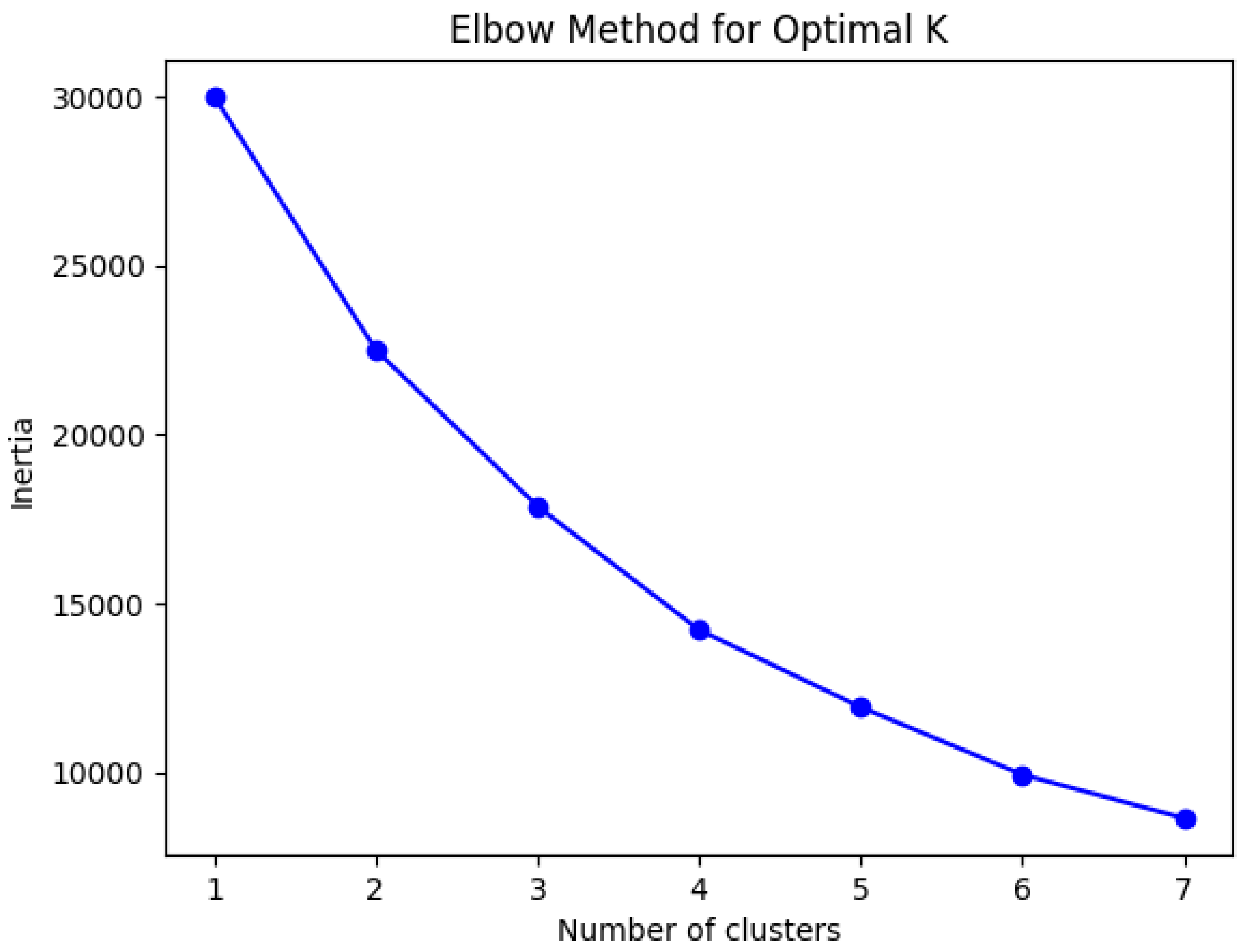
Results

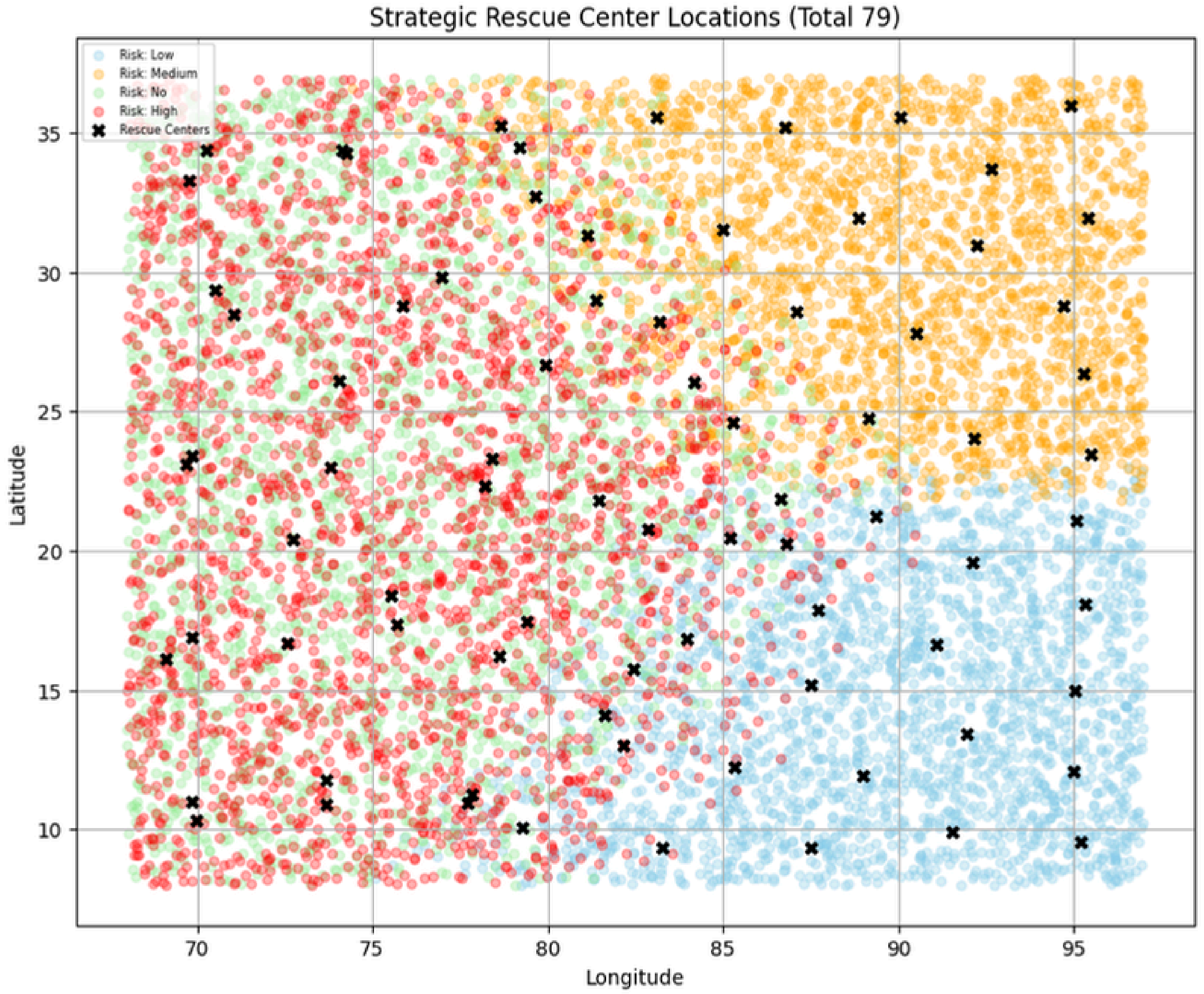


Latitude	Longitude	Water Level (m)	cluster	risk_label
18.861663	78.835584	7.415552	3	High
35.570715	77.654451	8.811019	3	High
29.227824	73.108463	4.631799	0	No
25.361096	85.610733	2.891787	2	Medium
12.524541	81.822101	3.188466	1	Low

Geospatial points clustered into 4 clusters using K-means, according to proximity and water level, and the risk labeled as ‘High’ , ‘Medium’, ‘No’, and ‘Low’.

Results

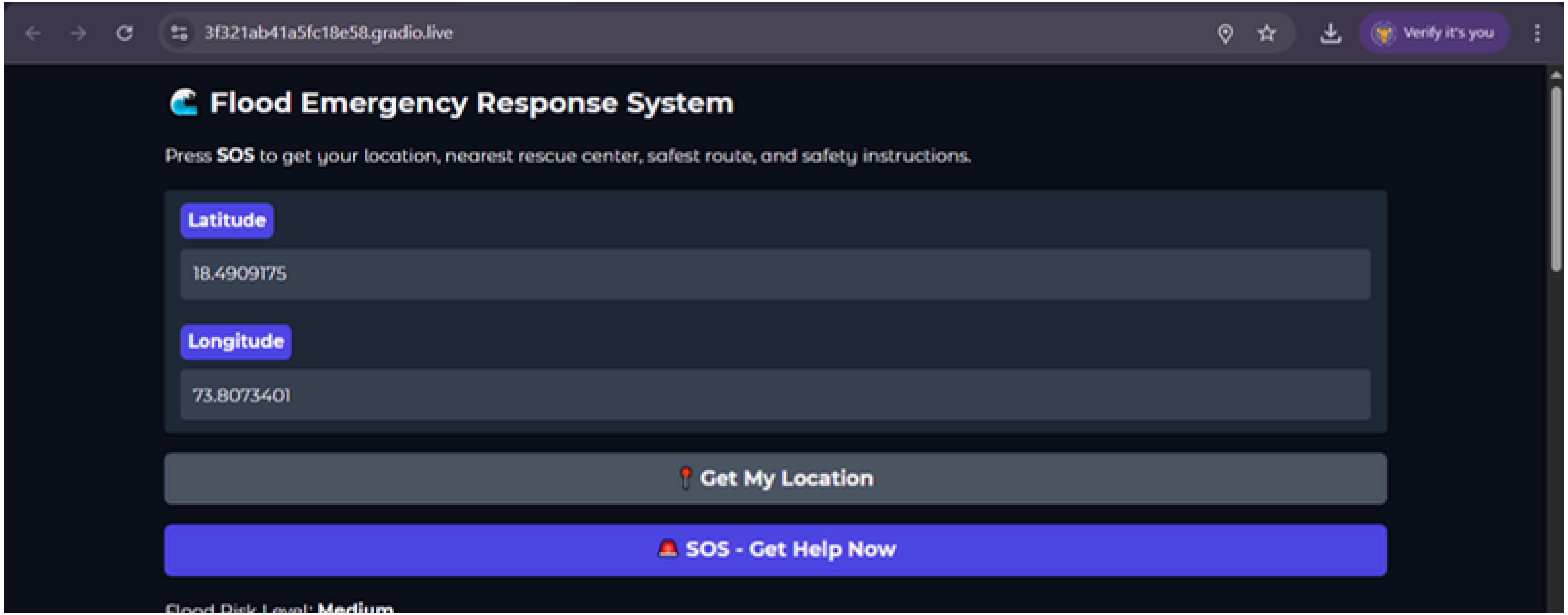




The elbow plot shows SSE/inertia dropping sharply until K=4, after which the curve flattens significantly. This indicates that K=4 is the optimal number of clusters.

Visual representation of strategically placed rescue centers, maintaining a minimum separation distance of 0.15 degrees of latitude/longitude between any two of them.

User Interface



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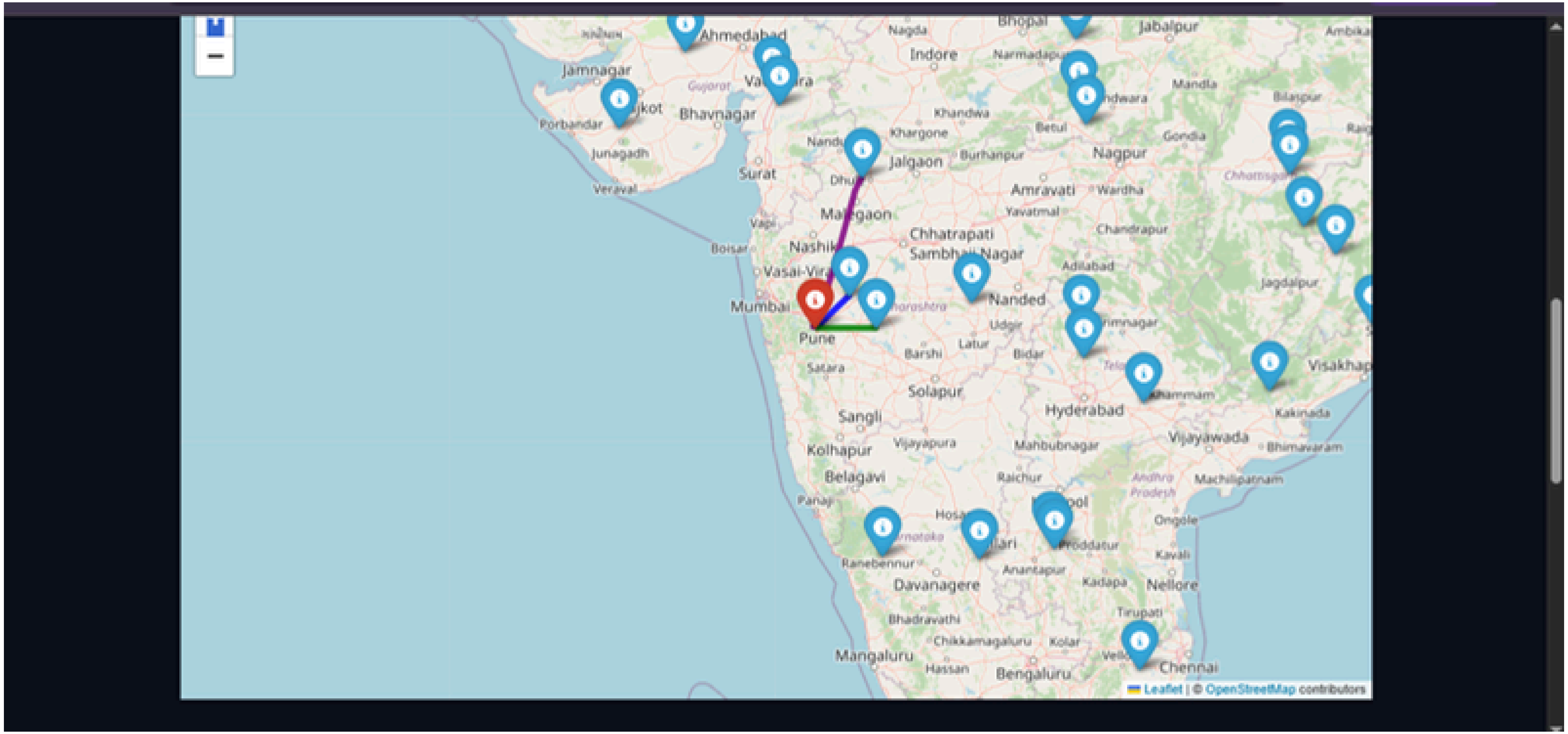

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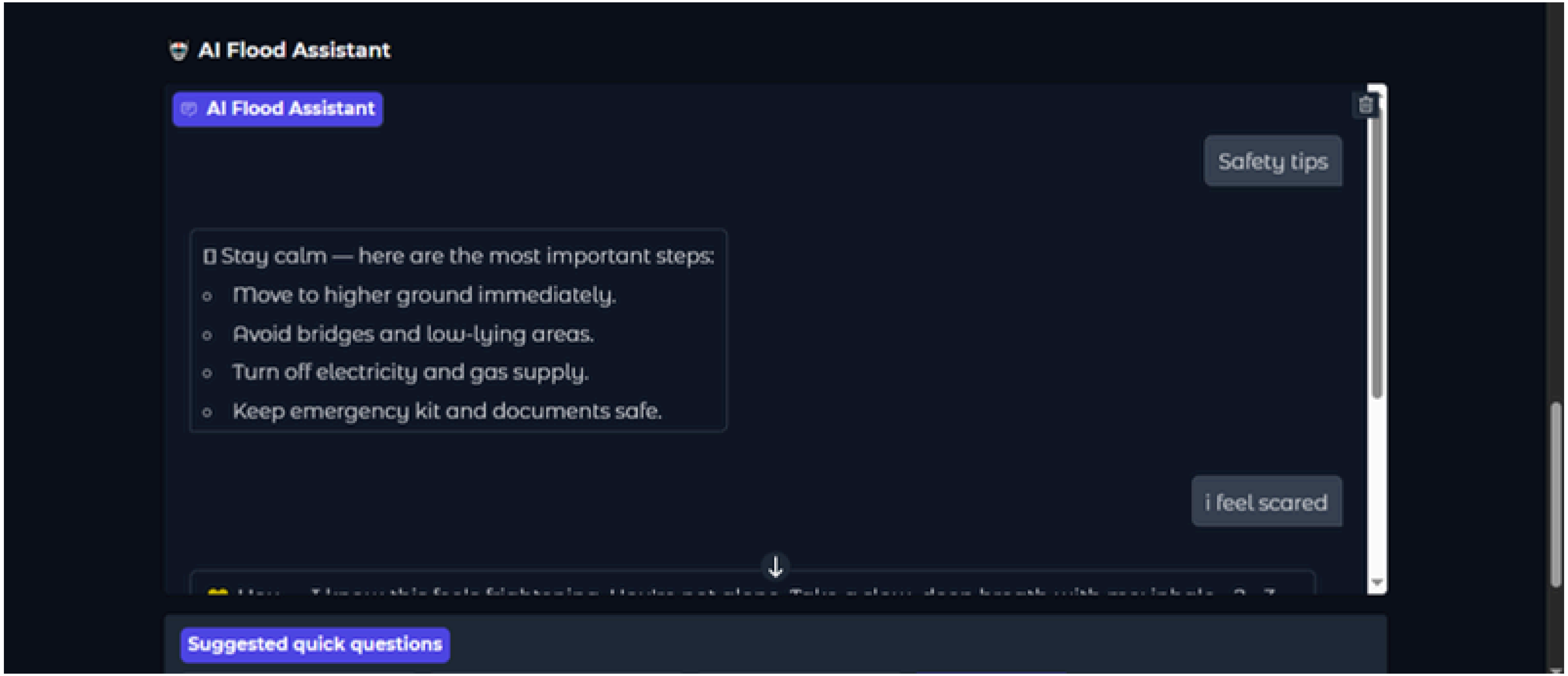
Chat Bot



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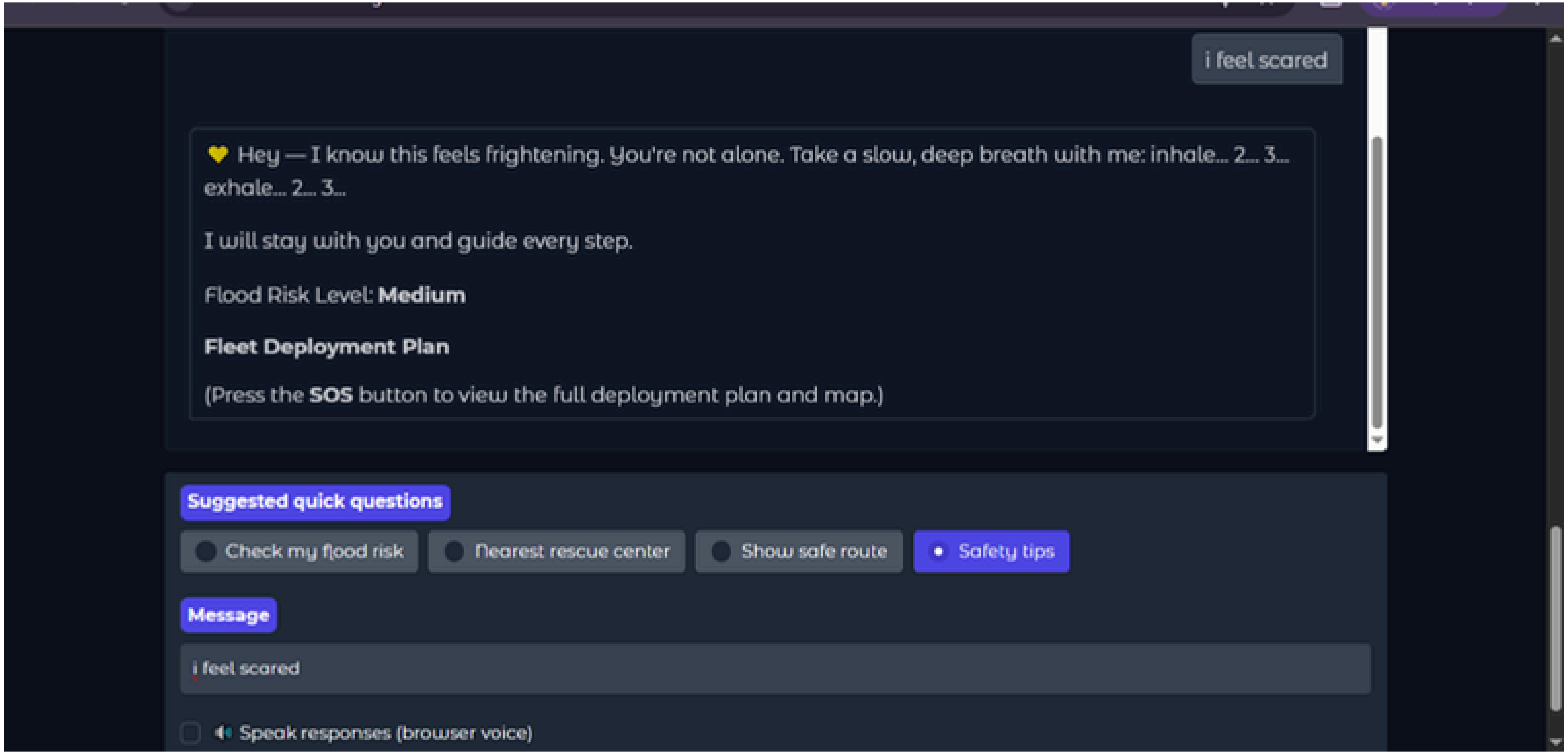

Discussion

Feature	Existing Systems	Our Flood Rescue System
Primary Focus	Rescue-Team Dispatch (Ref [8], [2])	Victim Self-Evacuation
Risk Assessment	Solely Shortest-Path / Distance (Ref [8])	Risk-Aware Decision-Making (13+ Flood Variables)
Data Model	Dependent on Fixed, Costly IoT Sensors (Ref [9])	**Data-Driven Risk Model** (No Hardware Dependency)
Impact	Creates delays; limited coverage.	**Scalable, Cheaper, Faster** deployment; empowers immediate action.

Technical edge: Risk-aware Routing Logic

The Problem with Current Routing:

- Classical algorithms (Dijkstra's, A*) prioritize travel-time minimization.
- Routing is based on shortest distance or estimated travel speed (Ref [2]).
- Crucial Flaw: They ignore real-world hazards, meaning the shortest path may be unsafe due to high water logging or rising flood levels.

Our Modified A* Algorithm	Advantage
Hybrid Cost Function	Integrates both Distance and Flood-Risk Intensity into the route cost.
Goal	Produces the safest route, not just the shortest.
Benefit	Ensures users avoid high-risk areas, making the route planning appropriate for real disaster conditions and improving survival chances.
Overall	The only system prioritizing route safety for direct victim self-evacuation.

Conclusion

- The system successfully integrates K-Means clustering, Logistic Regression, and a risk-aware A* algorithm to create an intelligent, real-time flood emergency response solution.
- It accurately identifies flood-risk zones, predicts risk for new locations, and provides safe, optimized evacuation routes.
- The Gradio-based interface enables instant SOS support, real-time risk assessment, and interactive route visualization for users.
- Overall, the project demonstrates a scalable, AI-driven framework that enhances public safety, supports authorities, and strengthens disaster management efficiency.

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Thank you !!